

Challenge 2 Encoder Decoder Key

```
message = input("Enter a message to encode or decode: ") # get a message
key = eval(input("Enter a key value from 1-26: ")) # get a key
message = message.upper() # make it all UPPERCASE :)
output = "" # create an empty string to hold output
for letter in message: # loop through each letter of the message
    if letter.isupper(): # if the letter is in the alphabet (A-Z)
        value = ord(letter) + key # shift the letter value up by key
        letter = chr(value) # turn the value back into a letter
    if not letter.isupper(): # check to see if we shifted too far
        value -= 26 # if we did, wrap it back around Z->A
    letter = chr(value) # by subtracting 26 from the letter value
    output += letter # add the letter to our output string
print ("Output message: ", output) # output our coded/decoded message
```